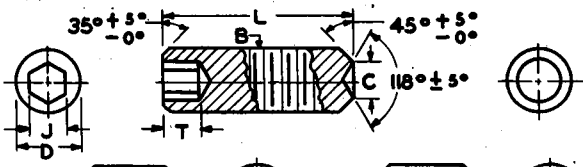
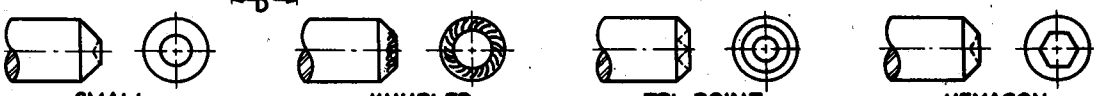
FED. SUP. CLASS.
5305





ACCEPTABLE CUP POINT DESIGNS

		#2 (.086)	#4 (.112)	#6 (.138)	#8 (.164)	#10 (.190)
D	NOMINAL SIZE					
B	THREADS PER INCH	56NC-3A	40NC-3A	32NC-3A	32NC-3A	24NC-3A
J	WIDTH OF SOCKET ACROSS FLAT — MAX	.0355	.051	.0635	.0791	.0947
	MIN	.035	.050	.0625	.0781	.0937
C	DIAMETER OF POINT — MAX	.047	.061	.074	.087	.102
	MIN	.039	.051	.064	.076	.088
T	KEY ENGAGEMENT MIN	.028	.040	.050	.062	.075

L LENGTH	TOLERANCE	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN	DASH NO.	FIIN
1/8		1		9		20		33			
3/16		2		10		21		34		46	
1/4	± 1-1/2% OF L WITH A MINIMUM OF ± .010 IN.	3		11		22		35		47	
5/16				12		23		36		48	
3/8				13		24		37		49	
7/16				14		25		38		50	
1/2				15		26		39		51	
5/8				16		27		40		52	
3/4						28		41		53	
7/8						29		42		54	
1						30		43		55	
1-1/4										56	

KEY ENGAGEMENT: THESE DIMENSIONS APPLY TO SCREWS ONE DIAMETER IN LENGTH OR LONGER. FOR SCREWS SHORTER THAN ONE DIAMETER IN LENGTH, SOCKET TO BE AS DEEP AS PRACTICABLE.
MATERIAL: STEEL, HIGH-GRADE ALLOY, IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.
HARDNESS: SETSCREWS SHALL BE HARDENED BY HEAT TREATMENT TO A HARDNESS OF ROCKWELL C45 TO 53.
PROTECTIVE COATING: PHOSPHATE COATED, IN ACCORDANCE WITH MIL-C-16232, TYPE II.
THREADS: THE THREADS SHALL BE IN ACCORDANCE WITH SCREW-THREAD STANDARDS FOR FEDERAL SERVICES HANDBOOK H-28.
DIMENSIONS: ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.
PART NUMBER: THE MS PART NUMBER CONSISTS OF THE MS NUMBER, PLUS THE DASH NUMBER. EXAMPLE: MS51019-1.
NOTES: (1) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON THE DATE OF INVITATIONS FOR BID.
(2) THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.

THIS DOCUMENT HAS BEEN PROMULGATED BY THE DEPARTMENT OF DEFENSE AS THE MILITARY STANDARD TO LIMIT THE SELECTION OF THE ITEM, PRODUCT, OR DESIGN COVERED HEREIN IN ENGINEERING, DESIGN AND PROCUREMENT. THIS STANDARD SHALL BECOME EFFECTIVE NOT LATER THAN 90 DAYS AFTER THE LATEST DATE OF APPROVAL SHOWN.

APPROVED 7 MAY 1958 REVISED

CUSTODIANS	OTHER INT.
A - ORD	A-CESIGT
N - SHIPS	N-AMCOR
AF - USAF	AF-

MILITARY STANDARD

SETSCREW, HEXAGON SOCKET, CUP POINT, ALLOY STEEL, PHOSPHATE COATED, NC-3A AND UNC-3A

MS51019

PROCUREMENT SPECIFICATION: FF-S-103
SUPERSEDES:
SHEET 1 OF 2

DD FORM 1 OCT 52 672-1

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D	NOMINAL SIZE	1/4	5/16	3/8	7/16	1/2
B	THREADS PER INCH	20UNC-3A	18UNC-3A	16UNC-3A	14UNC-3A	13UNC-3A
J	WIDTH OF SOCKET ACROSS FLAT MAX	.127	.1582	.1895	.2207	.252
	MIN	.125	.1562	.1875	.2187	.250
C	DIAMETER OF POINTS MAX	.132	.172	.212	.252	.291
	MIN	.118	.156	.194	.232	.270
T	KEY ENGAGEMENT MIN	.100	.125	.150	.175	.200

L LENGTH	TOLERANCE	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN
3/16	± 1-1/2% OF L WITH A MINIMUM OF ± .010 IN.	62									
1/4		63		81		99					
5/16		64		82		100					
3/8		65		83		101		118		135	
7/16		66		84		102		119		136	
1/2		67		85		103		120		137	
5/8		68		86		104		121		138	
3/4		69		87		105		122		139	
7/8		70		88		106		123		140	
1		71		89		107		124		141	
1-1/4		72		90		108		125		142	
1-1/2		73		91		109		126		143	
1-3/4		74		92		110		127		144	
2		75		93		111		128		145	
2-1/2						112		129		146	
3										147	

D	NOMINAL SIZE	5/8	3/4	7/8	1	
B	THREADS PER INCH	11UNC-3A	10UNC-3A	9UNC-3A	8UNC-3A	
J	WIDTH OF SOCKET ACROSS FLAT MAX	.3155	.378	.503	.5655	
	MIN	.3125	.375	.500	.5625	
C	DIAMETER OF POINT MAX	.371	.450	.530	.609	
	MIN	.347	.425	.502	.579	
T	KEY ENGAGEMENT MIN	.250	.300	.400	.450	

L LENGTH	TOLERANCE	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN
1/2	± 1-1/2% OF L WITH A MINIMUM OF ± .010 IN.	153									
5/8		154		169							
3/4		155		170							
7/8		156		171		184					
1		157		172		185		197			
1-1/4		158		173		186		198			
1-1/2		159		174		187		199			
1-3/4		160		175		188		200			
2		161		176		189		201			
2-1/2	± 1/32	162		177		190		202			
3		163		178		191		203			

NOTE: FOR MATERIAL, PROTECTIVE COATING AND OTHER PERTINENT DATA, SEE SHEET 1.

REVISED

APPROVED 7 MAY 1958

CUSTODIANS
A - ORD
N - SHIPS
AF - USAF

OTHER INT.
A- CESIGT
N- AMCOR
AF-

PROCUREMENT SPECIFICATION
FF-S-103

MILITARY STANDARD

SETSCREW, HEXAGON SOCKET, CUP POINT, ALLOY STEEL, PHOSPHATE COATED, NC-3A AND UNC-3A

SUPERSEDES:

MS51019

SHEET 2 OF 2

DD FORM 1 OCT 52 672-1

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